Applications Optimization Methodology—An Approach

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Abstract

Application development teams should take a proactive approach to optimizing the performance of their mission-critical applications. This paper describes the methodology successfully utilized by the Banc One Services Corporation to optimize mainframe business applications.

An Application Optimization (AO) team was formed to address this issue. Initially, the team conducted a two month planning phase. This phase defined the project scope, goals, objectives, project approach, team organization, roles, responsibilities, risk analysis, business success criteria, assumptions and project plan. The next phase, analysis, utilized a time box approach to identify areas of performance enhancement. This allowed the top ten business application systems to be optimized in a time frame required by the target completion date.

Various methodology techniques such as the association diagram, data gathering by interview, process definition, decision matrix, change control management and issue management, contributed to the success of this project.

The Applications Optimization project identified 1.3 million dollars in projected savings for the bank for a one year time period. Disciplined project management practices and methodology techniques are key factors in the success of optimizing business application systems.

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1.0 Introduction

In the spring of 1996, Banc One Services Corporation identified an immediate need to optimize the batch processing and online response times of their mission critical applications. With an expanding customer base and the need to merge existing customer application processing, the corporation found that the window of time available for batch processing was shrinking.

Initially management from the technical systems function was charged with reducing batch window processing time. There had been several SWAT team efforts, led by the technical systems function, in the recent past, that had identified and implemented various system performance tuning efforts. However, there still was a need to reduce batch processing time further, so upper management decided the appropriate approach was to focus on reducing applications' demand for systems resources. What was needed were ways to:

- reduce batch processing time (and increase the batch processing window),
- reduce CPU consumption, and
- reduce costs.

This would help the bank eliminate hardware purchases, reduce media requirements, reduce CPU requirements, and help to ensure that the nightly batch window would be met.

Due to the lack of communication between the technical systems function and application development function, the bank decided to form a dedicated Application Optimization (AO) team to supply a critical, missing link. The AO team provided value by understanding the technical environment and understanding how the business applications operate within it. They also understood the business application environment, why business applications were constructed, and then combined their knowledge in these areas. This enabled them to focus on potential optimization opportunities in the most efficient manner.

In May of 1996, five technical analysts were hired to accomplish this project. Team members were hired with different technical expertise, including personnel with methodology and team management experience, as well as technical specialties such as CICS, IMS, DB2, and media. Each of these analysts also had extensive application development experience. In total, the group had over 67 years of IS experience.

2.0 Project Initiation

The approach used to develop the project plan was based on standard project management practices and the LBMS methodology in use at Bank One.

The project began with the Project Initiation Stage, or in other methodologies referred to as the Project Planning Stage. The overall objective of this stage was to define, at a high level:

- the reason for the team's existence,
- what the team intended to do,
- how the team intended to accomplish their objectives,
- costs and benefits, and
- expected milestones and completion dates.

The strategic vision developed in the planning stage was in alignment with the bank's business plans and needs. Primary focus was to improve the effectiveness of business applications by improving performance, reducing operating costs, improving reliability, increasing availability, and improving maintainability.

2.1 Applications Optimization Team Mission Statement

The first task was to define the mission of the applications optimization team. The purpose of a mission statement is to establish the basis for the existence of the project. The team's mission statement encompassed what the project was, what the project would accomplish and the overall values and principles of the project. The use of a mission statement ensured that all groups within the organization worked together to ensure the success of the organization.

"To improve the effectiveness of mainframe business applications through the optimization of those systems."

2.2 Project Goals

After the definition of the mission statement was complete, the goals in support of the project were defined. These were the broad themes or aspirations of the project. They also identified the direction the project would focus its efforts on. Following are the goals defined for the bank's project.

- Assist application development teams with the improvement of their batch processing windows and online response times to counter the impact of data volume growth and to reduce operation costs.

- Track and report cost savings (DASD, CPU, tape, vault, fiche, print) and batch window time savings of implemented performance improvement recommendations

- Develop into a competency center and provide appropriate communication of optimization opportunities, provide standards and training to application development teams.

- Continually identify opportunities through the assistance of technical systems teams, application development teams, and vendors. Document and analyze each opportunity to improve application performance and determine if it is within or beyond the project scope. If opportunities are beyond the project scope, they will be reviewed later and a decision will be made to determine if it will be included in the AO long term strategy.

2.3 Project Scope

The next task in the project initiation phase was definition of the project scope. The scope was a detailed definition of the purpose of the project and was used to ensure that both business clients and application development teams were clear about what would be accomplished. The scope clearly defined the project boundaries—what the project would cover and what it would not cover.

Diagrams, such as a context diagram, were used to help present the project boundaries. A context diagram represents the boundary of a system with respect to information flow. They define how a system relates to its environment in terms of information it receives from, and provides to, its environment. At the center of the context diagram is the system whose context and boundary is being defined. Chart 1 is an example of a context diagram.

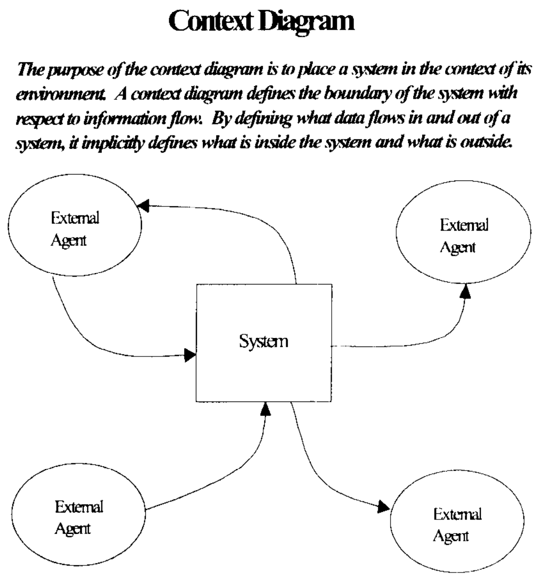

CHART 1. EXAMPLE CONTEXT DIAGRAM

As part of the project scope, the AO team identified eight technical optimization areas to focus on: media, IMS, CICS, DB2, JCL, job scheduling, long-running programs, and Pacbase. The team researched potential tuning recommendations in each area by interviewing application development teams, technical system teams, and vendors. Efforts were focused given the expertise of the team and the needs of the organization. Each team member was assigned a technical optimization area to analyze in detail for potential performance tuning recommendations. The group considered recommendations that would both reduce batch processing time and reduce CPU costs.

Each team member identified as many potential performance tuning recommendations as possible and then ranked each opportunity as to its type of benefit such as: CPU, I/O, elapsed time, tape, DASD, cost reduction. Each benefit also included a projection as to potential savings and difficulty of implementation (simple, moderate, difficult). This information helped determine whether a recommendation was worth the time to implement or not.

Following are some of the general optimization recommendations the team identified. Appendix A lists specific performance tuning recommendations for each opportunity area.

- Feasibility of BatchPipes implementation to reduce batch processing time. BatchPipes is a software product that allows jobs or steps that execute serially, due to data set dependencies, to execute simultaneously.

- Utilities were evaluated to determine efficiency. The team recommended using Favor as opposed to IDCAMS and SYNCSORT as opposed to DFSORT where possible.

- Using the *STROBE* MVS Application Performance Measurement System, the team evaluated long running programs. *STROBE* Performance Profile reports identified numerous areas to reduce processing time.

- Media usage was evaluated by identifying over-allocated files and supplying appropriate file allocation recommendations. Under-allocated data sets (data sets in 6 extents or more) were also evaluated and the team reviewed the use of AVGREC, STOP-X37, and numerous other JCL recommendations for these files.

- The AO team reviewed job schedules for ways to change the critical path of jobs to reduce batch window time. Some of the options were to remove unused reporting steps and reduce the number of backups.

2.4 Project Tracking and Reporting

To prove the benefits of the AO project, a Metrics and Reporting System was created. The purpose of this tool was to track and report cost savings of implemented optimization opportunities in a pictorial format. These included the application savings and cross applications impact savings.

2.5 Business Application Targets

After the team determined what potential optimization opportunities they should look for, they next determined which of the 150 business applications would be evaluated. This process began by defining a survey form used to interview application development teams. Since the main goal of our project was to reduce batch window time, this was the main criteria used to determine which business applications would be high priority.

The survey questions focused on printed output, application performance, and DASD usage. Each business application was rated as to their current situation. Application development teams were also asked to project future situations. This was important as the bank was in the process of consolidating business applications, and massive increases in data volume were expected in the near future. The rating devised for each question allowed a value from 1 to 5, 1 being no problem and 5 being extreme problems.

Computer operations and systems technicians were also interviewed to include their opinion of each of the applications. Mission-critical applications, as defined by the business, were given additional 50 points.

This information was then put into an association diagram (Applications Optimization Interview Matrix) that totaled the value for each business application. Information from operations and application development teams were combined to determine the total value. This technique was used to identify business applications with the potential to benefit the most from the Applications Optimization Team Analysis. The business applications were sorted in descending order by total. The top 10 business applications on this matrix were selected as those that had the most potential for benefit.

Chart 2, below, represents the bank's Applications Optimization matrix. In methodology terms it is referred to as an association diagram. Down the left side are the business applications and across the top are the different optimization opportunities. The total on the far right indicates the total rating for that business application. The business applications with the largest number indicated that they could potentially benefit the most from the optimization analysis.

Metric / Application	*Media*	*Printed Output*	*Batch Window*	*Operations*	*Priority Application*	Total
Appl 1	30	34	55	60	50	229
Appl 2	15	35	40	50	50	190
Appl 3	56	0	64	10	50	180
Appl 4	0	0	70	10	50	130
Appl 5	5	2	23	15	50	95
Appl 6	26	0	50	0	0	76
Appl 7	5	2	23	15	0	45

CHART 2. APPLICATION OPTIMIZATION MATRIX

2.6 Project Execution Plan

After the team targeted business applications for analysis, they then focused on defining the steps necessary for successful project execution. This phase of the overall project plan began with an analysis scenario developed in conjunction with the application development teams for the selected applications.

Appendix B lists the outline for each initial meeting with the selected application development teams. The purpose of this outline was to ensure that our goals were met at this meeting within a one hour time period. The goal of these initial meetings were to establish roles, responsibilities and expectations. The AO team role was to analyze the application and identify specific performance improvement recommendations based on the eight optimization areas. Within their analysis they were to project dollar and batch window savings. The application development teams were expected to assist with the analysis by providing expertise in regard to their specific business application.

2.7 Project Charter

The main deliverable from the Project Initiation stage was the Project Charter. This was considered to be a living document and was changed numerous times as the project evolved.

The primary benefit of a formal written charter is to focus the project and to prevent "scope creep", which inhibits the ability to reach project completion. All proposed project changes were required to go through a formal change management process. This process determined the impact of the intended change. Based on the impact, management decided whether to change or not change the scope of the project.

The charter consisted of the project mission statement, goals and objectives, high level executive summary, project scope, business success criteria, cost/benefit analysis, assumptions, roles and responsibilities, and the project execution steps and milestone dates. Once completed, the charter was then presented to upper management for approval. This document gave the AO team an outline of what needed to be done to be considered a successful project.

3.0 Project Execution

3.1 Application Analysis

The time box management concept was used to analyze each of the selected applications. A time box is a fixed schedule amount of time in which team members work to create a product. Time is the overriding factor and cost and scope may be adjusted to fit within the time box.

2.5 weeks were allocated to complete the analysis of each application. Each AO team member, based on their technical expertise, analyzed the applications using the optimization opportunities that were identified in the planning stage. The team looked for potential improvements that would provide the most benefit for the least amount of effort. The AO team focused first on each application's daily critical-path jobs. Their next priority were critical path monthly jobs and if any timed remained, they team focused on non-critical-path long running jobs.

The AO team was divided into 4 work groups: one to analyze utilities and long running programs (using *STROBE*), one to analyze IMS, one to analyze CICS, and the remaining group to analyze JCL and scheduling.

Some of the groups' activities included the identification of the critical path jobs, reviewing JCL for potential removal of steps for backup, identification of reporting steps to be taken out of the critical path, review of file allocations, review of file buffering, identification of more efficient utilities and numerous other optimization opportunities as had been identified at the beginning of the project.

3.2 Recommended Changes and Benefits Report

As a result of the application analysis, the AO team created a report called "Recommended Changes and Benefits Report." This report defined the specific recommendations that needed to be made to improve the performance of the business application. Each recommendation included an evaluation as to difficulty of implementation and benefit of implementation. Benefits were defined in terms of measurement type such as: CPU, DASD, elapse time, response time, I/O, etc. The report also identified how to make the change and sometimes included samples.

3.3 Implementation of Recommended Changes

Upon completion of the Recommended Changes and Benefits Report, a meeting was held with the application development teams to review the recommendations resulting from the application analysis. Because the application development teams were the experts on the business applications, they needed to evaluate each of the potential improvements to determine if they made sense for their specific business application. It was the responsibility of the application development teams to test the recommendations and make the final implementation determination.

When there was a need, the AO team also scheduled subsequent meetings for more in depth review of some of the recommendations. Approximately two weeks following the initial review, application development teams were expected to develop a plan for implementation of identified opportunities. The AO team assisted with this process.

A spread sheet was developed placing high priority projects first. Implementation plans included expected implementation dates, benefits of implementations, and reasons for not implementing any of the recommendations. The total benefit in dollars, window time savings, CPU reduction, EXCP reduction and response time reduction were summarized at the end of the plan.

The AO team continued to offer support to these application development teams and also monitored the implementation of the changes to determine the actual benefits of the optimization opportunity recommendations.. They viewed the application development teams as co-suppliers in the process and were committed to their success. While the AO team monitored and supported the application development teams with the implementation of the identified opportunities, the application development teams performed most of the actual code changes and testing. The AO team was expanded, adding two consultant programmers to assist application development teams with the implementation of the recommendations when they were too busy to address the changes. This however, was not the main focus of the AO team efforts.

Support for the other 140 mainframe application development teams not included in the application analysis was provided by communicating the optimization opportunities identified. The AO Team also provided training as needed in regard to some of the identified opportunities.

3.4 Project Results

In the first five months of the project, the cost of the AO team, (5 technical analysts for 5 months at $70,000 per year) was a cost of $147,000. After analyzing the first three applications, the team identified a projected benefit for a twelve month period of $1,300,000. This was an annual yearly projected dollar cost avoidance. Projected reduction in daily batch window time for the first three applications amounted to 250 minutes. The projected monthly reduction in batch window time for these applications came to 550 minutes. These numbers were just for the first three applications of the ten to be analyzed. Appendix C lists the AO project total projected benefits.

The first application analysis was completed with the assistance of the application development team. The AO team made a number of recommendations with estimated savings of $80,000 from reduced CPU, DASD and response time. A specific recommendation was to reduce over allocation of sequential DASD files that were then over allocated on average of 600% per day. The team also identified a potential reduction of 43 minutes of batch window time for daily processing. The majority of the time savings would be a consolidation of reports as identified by application development team. A projected 314 minute reduction for monthly processing was identified.

In the analysis of the second application, the team made recommendations with estimated savings of $70,000 from reduced CPU, DASD and response time. The team projected a reduction of 25 minutes of batch daily window time.

The third application development team was assisted by the AO team and CICS system programmers to resolve problems with "LOAN" transactions. This transaction was responsible for approximately 10% of the bank's online CPU time. The team identified a continues loop that once corrected, amounted to 36,080 seconds of CPU time reduced per week. This amounted to savings of $10,000 per week or $42,808 for the one month period of time that this problem occurred.

The AO team facilitated the implementation of the PARDLI option for another application. This was an IMS option identified by the technical systems function. This was projected to reduce elapsed time and CPU by

50% of time specific to IMS calls. This opportunity addressed a critical processing issue that affected other business applications—a name and address retrieval function that obtained data from an IMS database. This had been a long running issue for many business applications. The AO team identified all applications impacted by this change and worked with the application development teams to implement the solution.

Another area the team participated in was the facilitation of the Microfiche Output Improvement Project. The AO team's role was to be the interface between application development teams and computer services. Phase one of this effort had been implemented and phase two was expected to be completed within several months. This effort was projected to save the bank over $600,000 in six months and also helped consolidate reporting bundles in preparation for future business application consolidations.

4.0 Conclusions

Banc One's successful application optimization project was due in large part to the structured methodologies and processes that were used. There are several recommendations for organizations undertaking similar projects.

1. Such a project must have commitment from top management, middle management and anyone participating in the effort. Participants will not put passionate effort to the project if they are not committed and lack of a motivated team will challenge the project to complete successfully.

2. Organizations should identify goals and objectives that are in line with their department and overall company strategy. An application optimization team's reason for existence must be in support of the company's business objectives. At Banc One, the AO team worked with application development teams and shared a common goal to ensure that their batch window times could be met as the bank continued to grow and consolidate. This was also in line with maintaining a strategic focus. It was critical that the team stayed focused on the bank's business plans and needs.

3. Staffing of an applications optimization team should be a key consideration. Initially, experts should be hired to give the team immediate credibility from both the technical systems function and the application development function. It is also important to the division of labor that specialists are hired. As the optimization team evolves into a competency center the staff will be viewed as technical consultants.

4. Another critical challenge is the management of expectations. Everyone involved in the project must clearly understand what they will be expected to do and what they can expect from the outcome of the project. At Banc One, the AO team clearly defined expectations with the clients at the outset of this project. This also included having a clear understanding of the AO team responsibilities and other organization responsibilities in support of our mutual clients.

5. It is extremely critical to have team members who are very committed to the technical details of the project. Communication skills are also very critical. It makes for a very long and challenging project if the project has "tech heads" that do not have any type of communication and conflict management skills. Technical team members need to have the ability to act as a consultant and pass on the information obtained in a format that can be understood. The application development teams need to understand the changes in order to change their systems. At Banc One, maintaining ongoing open communication with application development teams, computer services and management helped to ensure success.

6. In order to work effectively with the application development teams, it is crucial that a positive working relationship be developed. AO team members should consider themselves to be in "marketing" and should sell the benefits of their ideas to the overworked and extremely stressed application development teams. Application development teams in general are too busy to be bothered with performance projects.

7. Use the time box management concept to limit the time spent in analysis. The top 70% of performance issues can be identified in a short period of time. This eliminates the concept of analysis paralysis, or spending a lot of time on something that may provide little benefit. If work extends beyond the target dates, organizations should review the process to see if they can eliminate any low-yielding activities.

8. Stick with the project plan as developed and defined in the charter. If every neat project that management identifies is "attacked", the project will not accomplish anything of value because too many goals are in focus at the same time. Change management control should be implemented to revise the project schedule and scope.

9. Focus on project goals. It can not be stressed enough that organizations must maintain focus on what they planned to do. Projects of value will be completed if the focus is narrow enough. When team members are working on too many things at the same time they usually get lost and get nothing done. Narrow the path and complete the project. This also requires keeping management focused on what is in scope.

10. Measure the benefits of the implemented application performance improvement opportunities. Management will require proof of the benefits of an AO team. Baselines and measurement methodologies will need to be established in the beginning of the project. Then track and report the savings on a regular basis.

11. Market internally and celebrate success with your team!